

Overview of Machine Learning and Deep Learning

Content

- ✓ History and Evolution of AI and Deep Learning.
- ✓ Introduction to Machine learning.
- ✓ Difference between traditional programming approach and machine learning approach
- ✓ Types of Machine learning techniques
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement learning
- ✓ Introduction to Deep learning
- ✓ Difference between Machine Learning and Deep Learning
- ✓ Applications of Deep Learning

History of AI and Deep Learning

A Brief History of AI with Deep Learning

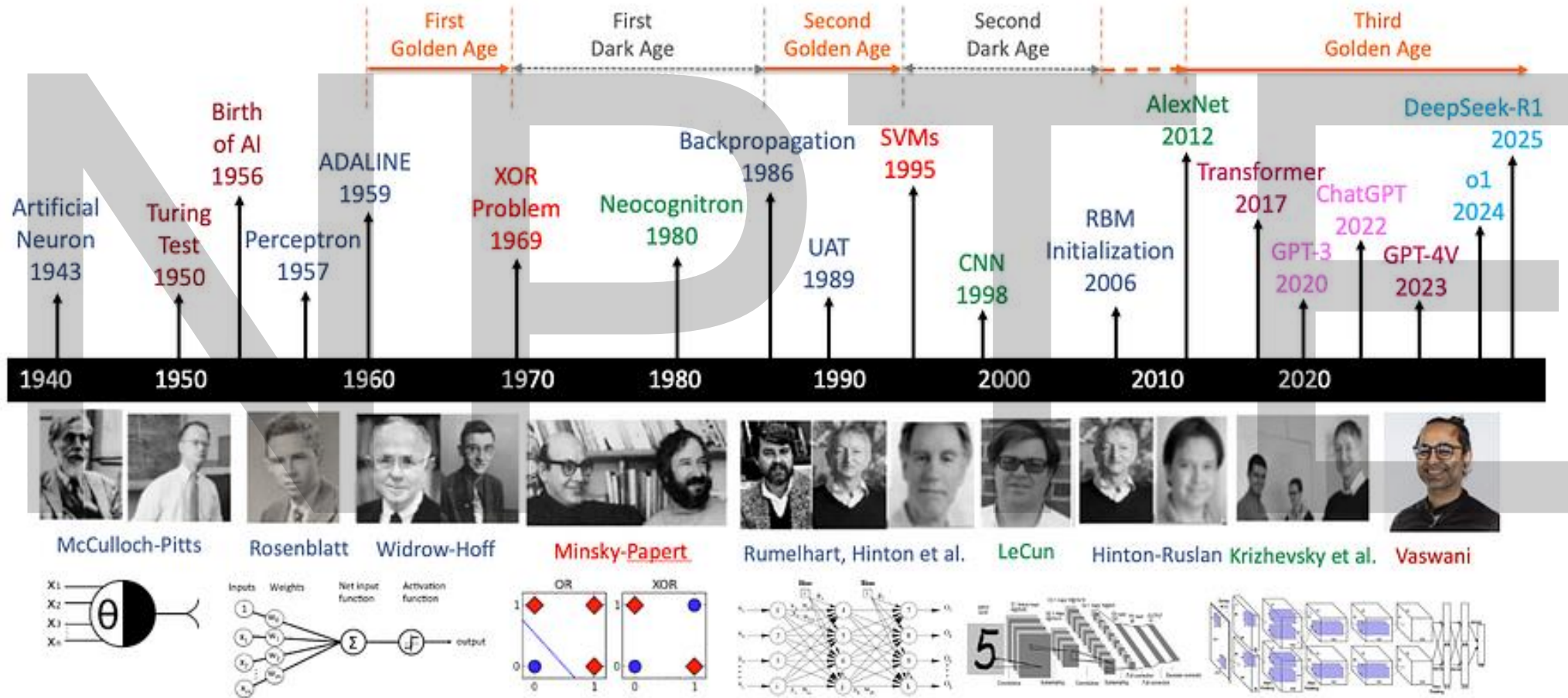


Image Ref:

<https://medium.com/@lmpo/a-brief-history-of-ai-with-deep-learning-26f7948bc87b>

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Computers

Era	Hardware	Key Features	Relevance to DL
1950s–80s	CPU (single core)	Sequential, low speed	Only basic AI algorithms possible
1990s–2000s	Multi-core CPUs	Parallelism began, but limited	Could train small models, slow
2010s	GPUs (NVIDIA CUDA)	Massively parallel, fast matrix ops	Huge boost to Deep Learning
2016–Now	TPUs / Specialized AI Chips	Designed for tensor ops, faster than GPUs	Cloud-scale training, real-time inference

Year	Hardware	Time to train ImageNet
2012	CPU	Weeks
2015	GPU (GTX 1080)	Days
2020	GPU (A100)	Hours
2024	TPUv4	Minutes

What is Machine Learning

- ML is a subset of AI that enables systems to learn from data.
- Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed.

Traditional Programming Approach

Problem statement: Whether a patient is having ovarian cancer or not?

Based on a set of fixed rules, like:

✓ If tumor size > 5 cm AND CA-125 level > 35 AND irregular shape on ultrasound $\rightarrow$ likely cancer.

- ✓ The symptoms and test results vary widely between patients.
- ✓ Writing **all possible rules** to detect it accurately is nearly impossible

Machine Learning Approach

Train a machine learning model using medical data from many patients:

- ✓ Imaging data (ultrasound, MRI)
- ✓ Blood test markers (like CA-125)
- ✓ Patient age
- ✓ Previous diagnosis(cancer or not cancer)

- Model learns patterns in the data that are common among patients who had cancer.
- When a new patient's data is given, the model can **predict the likelihood of cancer.**

Types of Machine Learning Technique

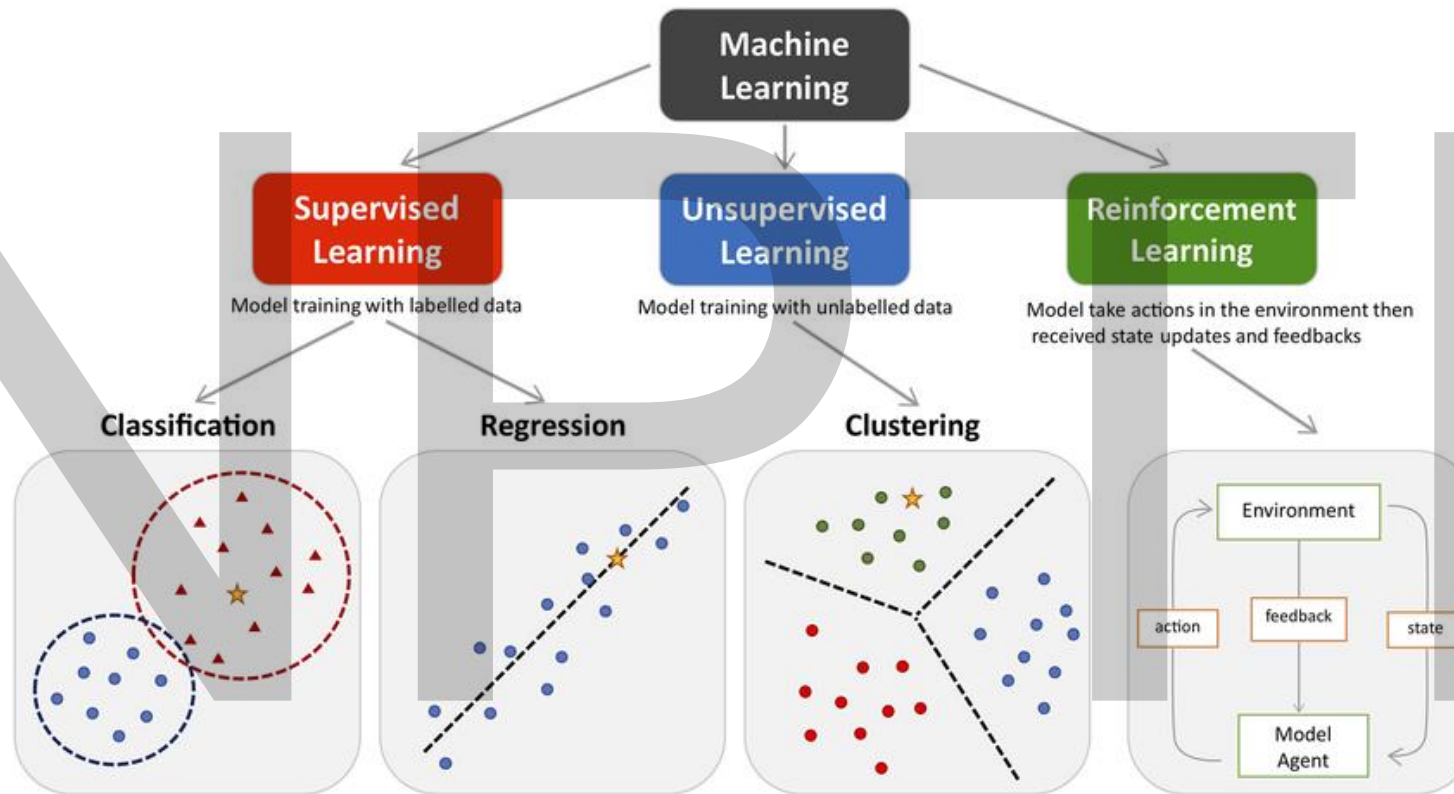
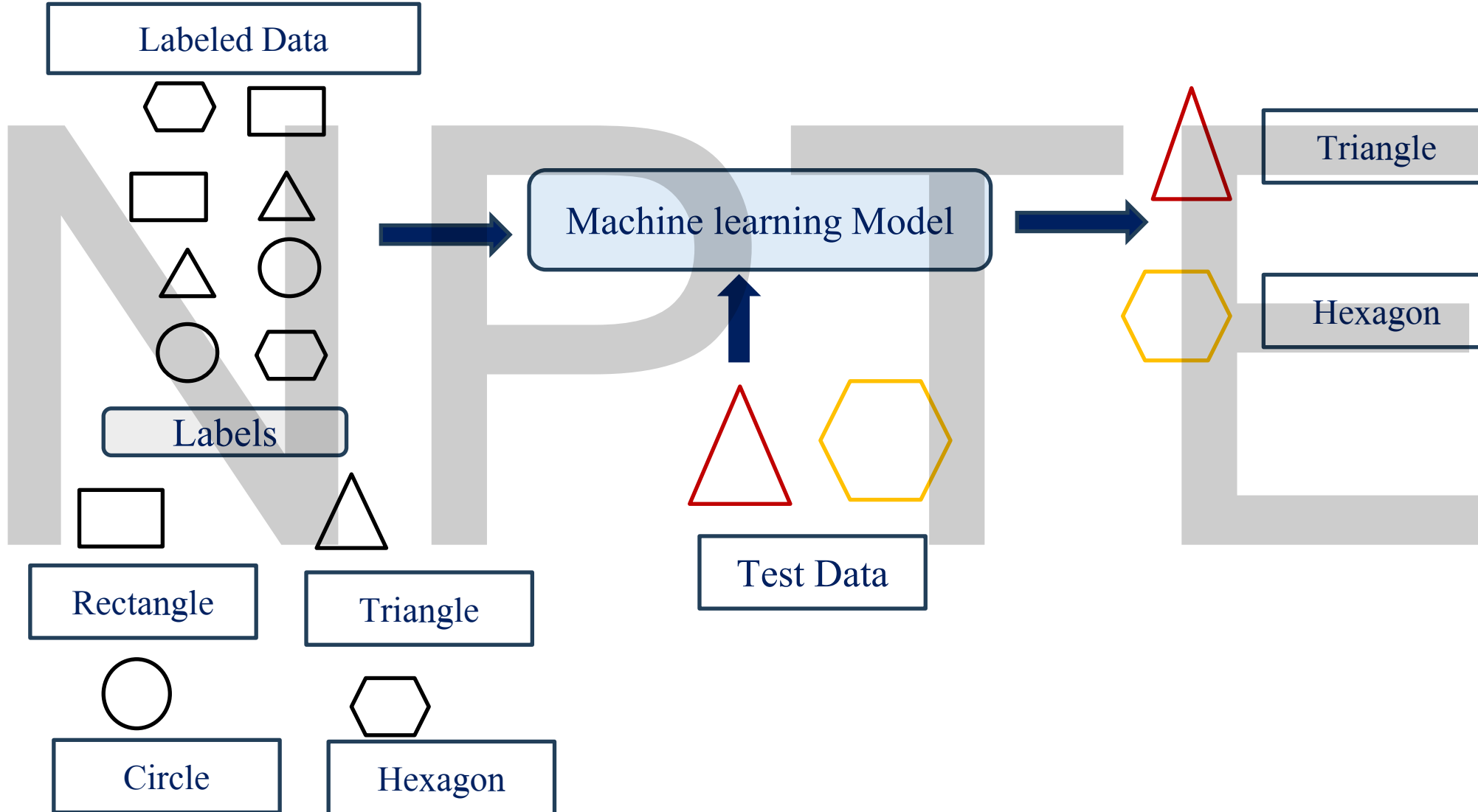
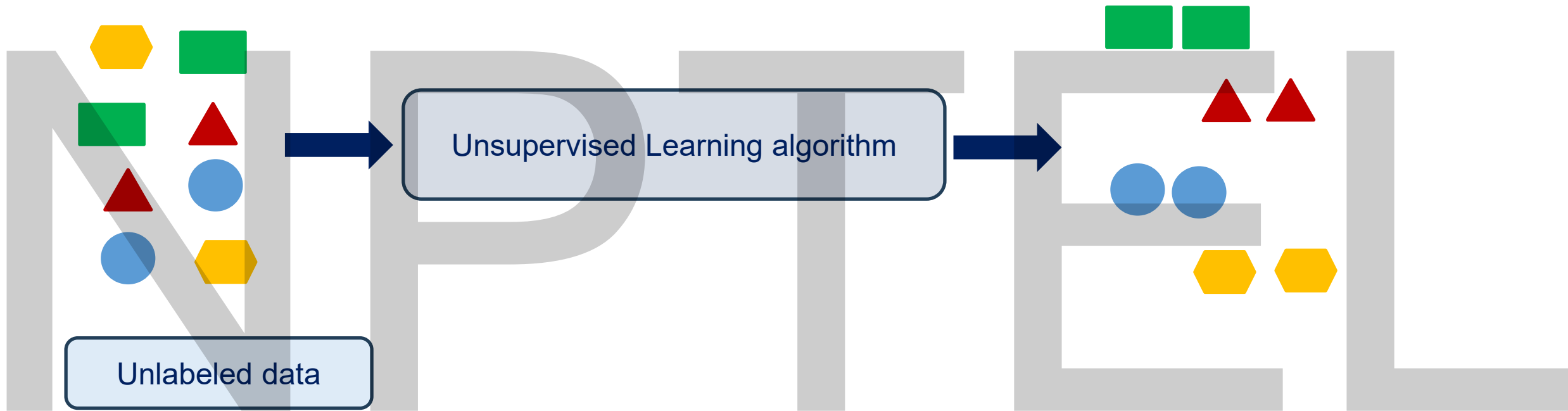


Image ref:
Peng, Junjie & Jury, Elizabeth & Dönnies, Pierre & Ciurtin, Coziana. (2021). Machine Learning Techniques for Personalised Medicine Approaches in Immune-Mediated Chronic Inflammatory Diseases: Applications and Challenges. *Frontiers in Pharmacology*. 12.

Supervised Learning



Unsupervised Learning



Reinforcement Learning

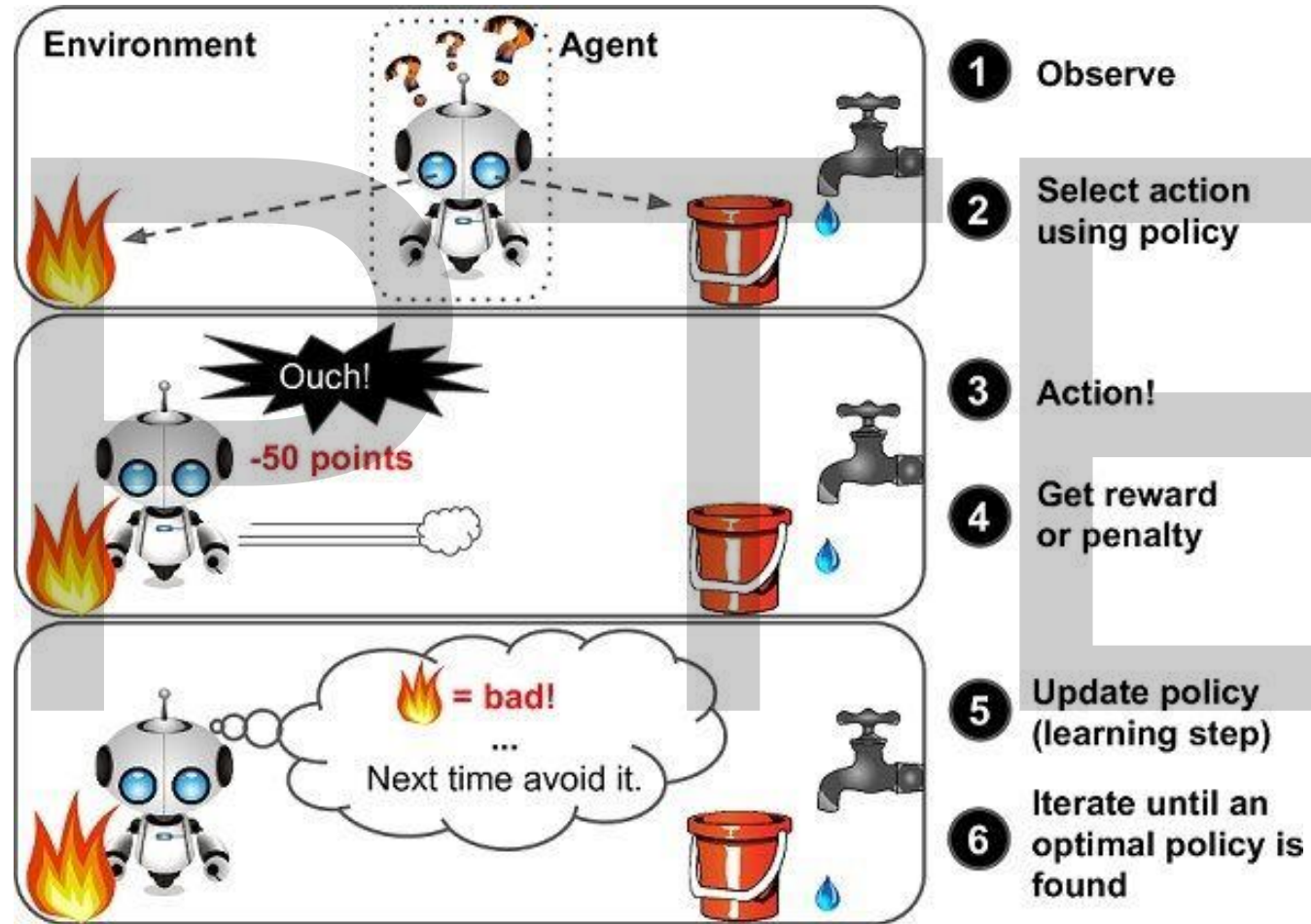
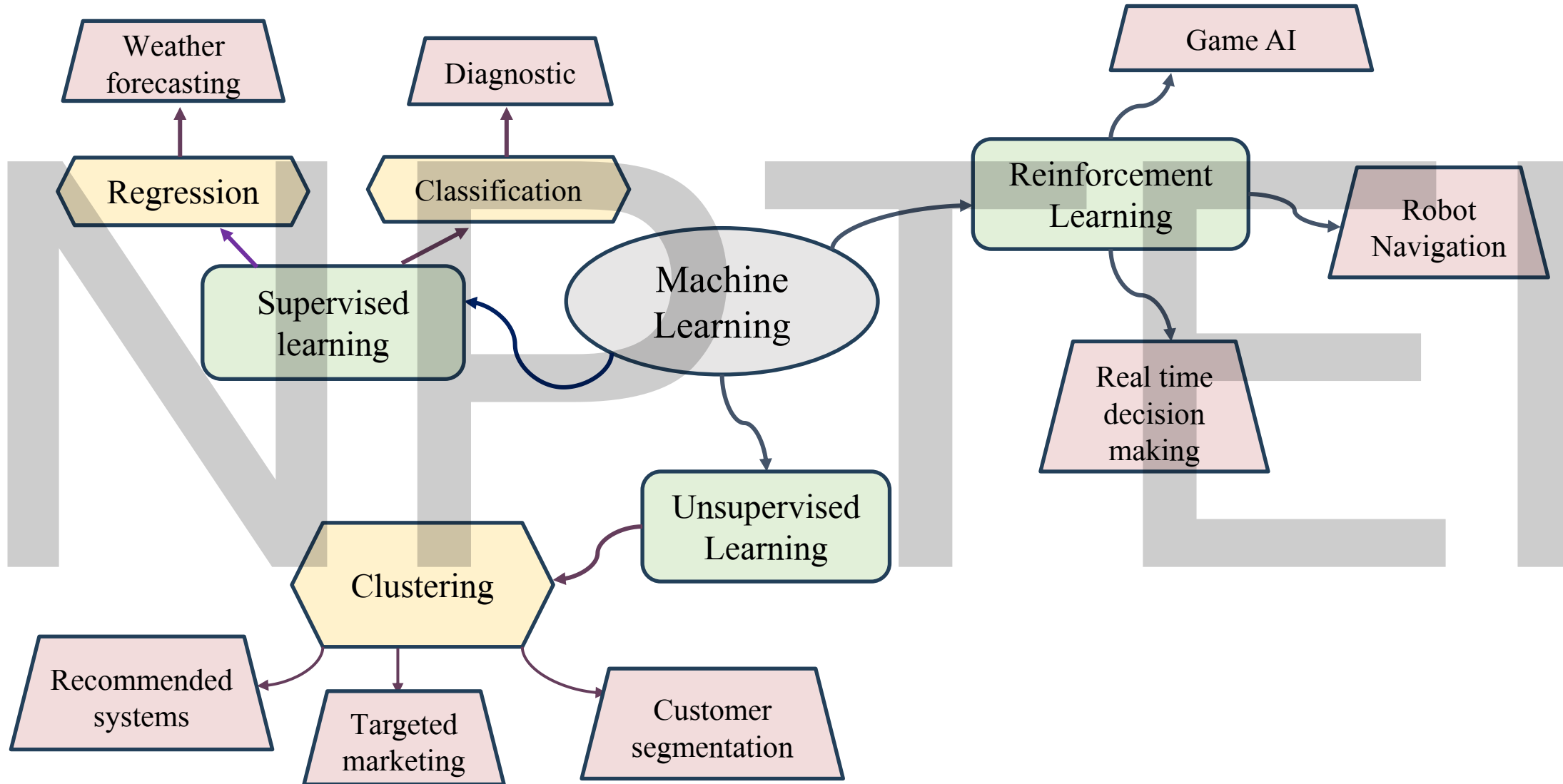


Image ref:
<https://marutitech.com/businesses-reinforcement-learning/>

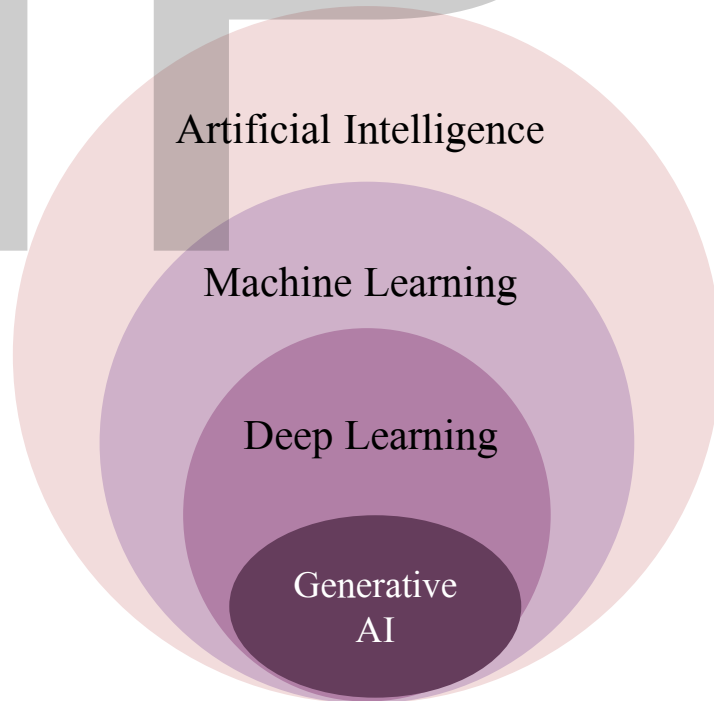
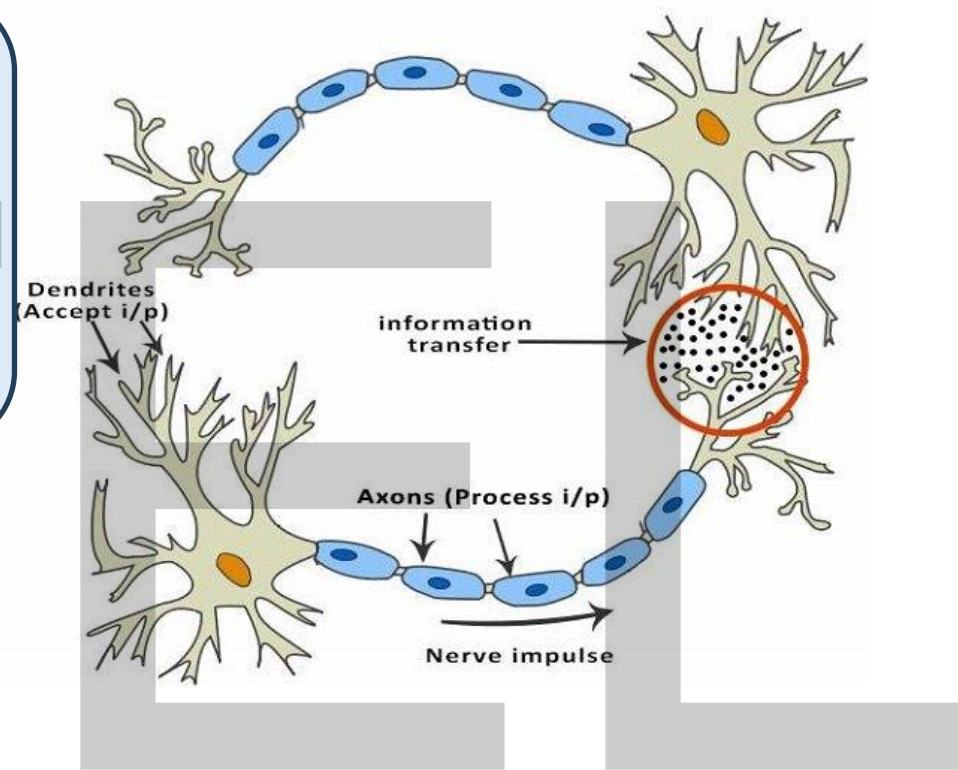
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Applications of Machine Learning



What is Deep Learning

- ✓ Subset of Machine learning based on artificial neural networks
- ✓ Inspired by the human brain
- ✓ Handles large and complex data (images, speech, text)



Difference between Machine Learning and Deep Learning

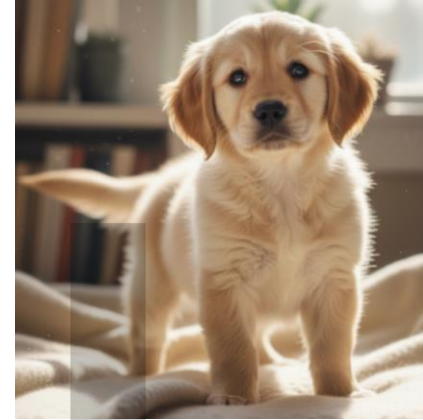
Problem statement: Image Classification

ML Approach

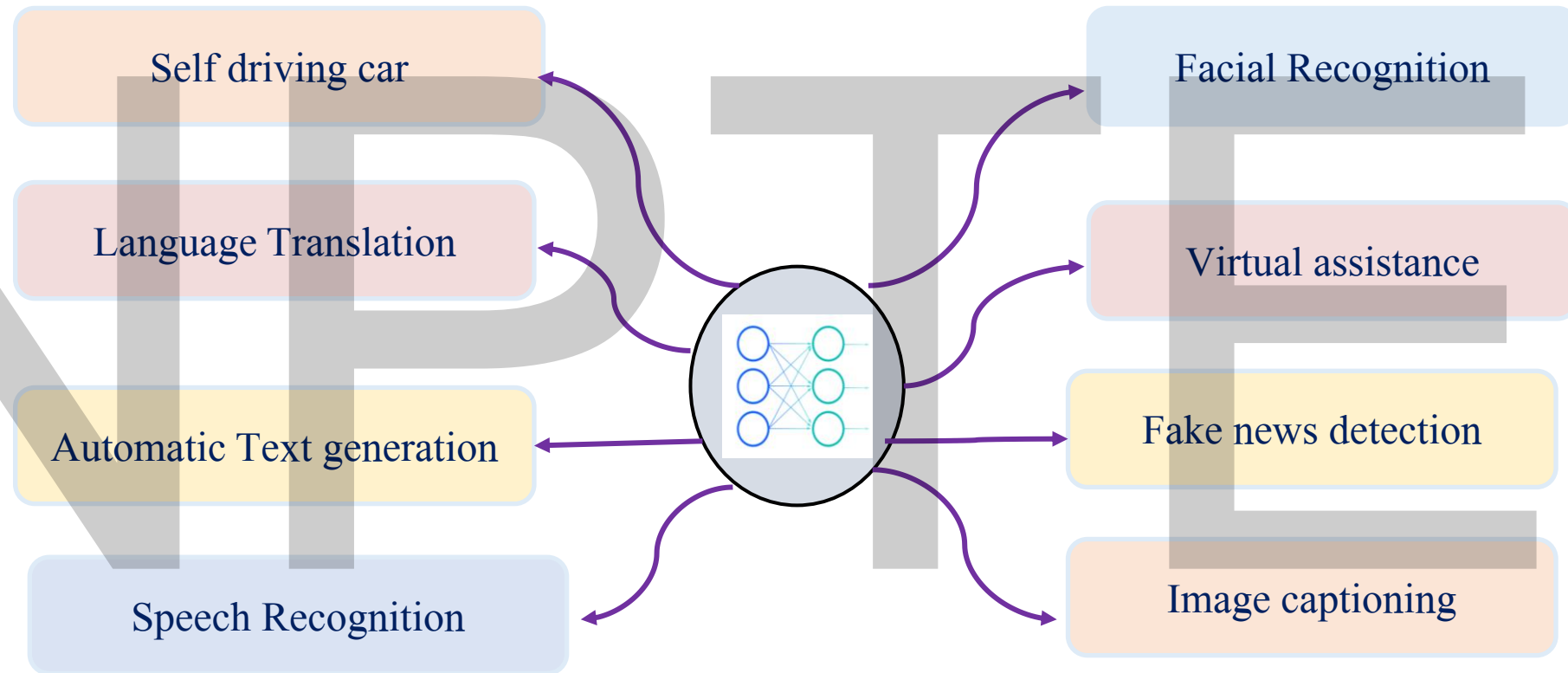
- ✓ Humans first **extract features** from the images:
 - Color histograms
 - Edge detection (shapes, outlines)
 - Texture patterns
- ✓ These features are fed into an algorithm like **Support Vector Machine (SVM)** or **Random Forest**.
- ✓ Performance depends on how well humans design the features.

DL Approach

- ✓ A Convolutional Neural Network (CNN) is trained directly on raw images (pixels).
- ✓ The network automatically learns hierarchical features:
 - First layers → detect edges
 - Middle layers → detect eyes, ears, fur patterns
 - Final layers → identify “cat” vs “dog”
- ✓ No need for manual feature extraction—model learns everything from data.



Applications of Deep Learning



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Summary

Next Session

Introduction to Single Layer Perceptrons

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Thank you